

Basic Details

Department: Computer Engineering

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Project Code: -

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Program: Final Year B. Tech

Course Name: Bachelor of Technology

Project Type: Major

Guide Details

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Designation: -

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Area of Expertise: -

Project Details**Description:**

NextGen Rakshak is a hybrid edge-based system engineered for the rapid recovery of lost children at mass gatherings such as festivals, fairs, and religious congregations (e.g. Kumbh Mela, Ganesh Chaturthi visarjan processions, Dussehra grounds, and urban temple fairs). The system is built around a three-component architecture in which the heavy computational work of face recognition happens on the edge, on ordinary volunteer smartphones, rather than on a central server, while a cloud backend coordinates alert distribution. This project directly proposed the "golden hour" problem repeatedly highlighted by child-rights experts: the first sixty to ninety minutes after a child goes missing in a crowd are the most critical for recovery, yet this is precisely when cellular networks at large events become congested and existing government portals (which are built for post-incident record-keeping rather than live search) are least useful.

Background & Need:

1. 1.1 The Scale of Missing Children in India India faces a persistent and deeply alarming crisis of missing children. The National Crime Records Bureau (NCRB) data for 2022 reveals that 83,350 children were reported missing in a single year — comprising 20,380 males, 62,946 females, and 24 transgender minors. Despite tracing efforts, a gap of over 2,700 children remained unbridged that year alone. Looking at the cumulative picture, over 47,000 children are currently listed as missing in India, with 71.4% being minor girls. The NCRB's 2023 figures record approximately 82,000 children listed as missing alongside 1,77,000 registered cases involving crimes against children. The trajectory over recent years shows a mostly rising trend: a 7.5% spike in 2022 compared to 2021, a significant 30.8% surge in 2021 against 2020. Between 2018 and 2023, 2.75 lakh children went missing, of whom 2.12 lakh were girls. Approximately 36,000 children remain untraceable since 2020, as the Centre informed the Supreme Court in February 2025. One of the most cited statistics— "one child goes missing every eight minutes in India" —was acknowledged by the Supreme Court itself, with Justice B.V. Nagarathna stating on 18 November 2025: "I have read in a newspaper that every eight minutes, a child goes missing in the country. I don't know if this is true or not. But this is a serious issue."
2. 1.2 The Recovery Gap Despite coordinated police efforts and multiple government portals, recovery rates remain troubling. A FACTLY analysis of NCRB data shows that in 2023, out of 8.68 lakh total missing persons reported, only 4.6 lakh were traced — a recovery rate of just 53%. For children specifically, NCRB's 2022 data indicates 80,561 children were traced or recovered out of 83,350 reported, leaving a gap each year. The government's TrackChild portal data shows that between January 2012 and June 2015, police uploaded information on 1,41,605 missing children but only 55,660 were traced after matching profiles. Former Delhi DGP Amod Kanth noted that nearly 25 to 30 per cent of missing persons are never found, calling it "extremely worrying" and identifying child labour, forced labour, and human trafficking as the strongest contributing factors.
3. 1.3 The Critical First Hours Both police officials and child rights activists emphasise that the first 24 hours are crucial to tracing a missing child. Former Delhi Police Commissioner S.N. Shrivastava stated that "tracing missing children should be done on priority" as these children are at immediate risk of being pushed into begging, sexual exploitation, or criminal networks. Rishi Kant of Shakti Vahini flagged "shoddy investigations in the first 24 hours" as a key reason for the persistently high number of unresolved cases. However, at large festivals and crowded events, the first hours are precisely when police are most overwhelmed and lack any real-time, automated mass-alert mechanism.

Problem Statement:

1. 1. The Scale of the National Crisis India reported 98,375 missing children in 2024 (NCRB), averaging 269 children per day. Despite a national recovery rate of ~67%, these recoveries often take months or years, leaving children vulnerable to trafficking, forced labour, and exploitation.
2. 2. The Critical "Golden Hour" Failure Child-rights experts and police officials unanimously emphasize that the first 60–90 minutes ("Golden Hour") are the most critical for tracing a missing child. At large mass gatherings (Kumbh Mela, Dussehra, Ganesh Chaturthi), police are severely overwhelmed during this window and lack any automated, real-time mass-alert mechanism to mobilize searchers.
3. 3. Technical Gaps in Existing Solutions All existing systems (TrackChild, Khoya-Paya, ReUnite, Project Guardian, Maha Kumbh AI centres) share fundamental limitations: Reactive Model: They operate on an "upload-first" basis (requiring someone to find and photograph the child), rather than enabling proactive crowd-scanning. Centralized Dependency: They rely on central cloud servers for face matching (e.g., Amazon Rekognition), causing latency and single points of failure. Internet Dependency: They fail during network congestion—which is the norm at India's largest festivals where cellular towers are completely saturated. Closed Networks: They are restricted to police personnel only, failing to leverage thousands of credible on-ground volunteers (NCC/NSS cadets, security staff, shopkeepers).
4. 4. The Operational Bottleneck A single police Lost & Found kiosk can manually scan only 50–100 faces per hour. In a festival crowd of 100,000+, this manual approach is functionally useless during the critical first hour, drastically reducing recovery chances.
5. 5. The Privacy & Infrastructure Challenge Existing cloud-based face-recognition solutions raise significant privacy concerns by uploading sensitive biometric data to central servers. India's unique festival infrastructure lacks reliable internet connectivity, making existing centralized architectures inherently unsuitable for deployment.

Objectives:

1. 1. To design and develop a hybrid edge-based system for rapid lost-child recovery at mass gatherings, addressing the critical "golden hour" gap identified by child-rights activists and law-enforcement experts.
2. 2. To implement on-device face detection using Google ML Kit and face recognition using MobileFaceNet with TensorFlow Lite, achieving real-time performance on standard Android devices without any cloud round-trip.
3. 3. To enable both internet-based push notifications (via Firebase Cloud Messaging) and a fully offline, custom multi-hop store-and-forward mesh built on the Google Nearby Connections API (Bluetooth LE for discovery, Wi-Fi Direct for high-speed transfer) for alert distribution, so the system keeps working when cellular networks are congested.
4. 4. To develop a police kiosk web application using Next.js with Firebase integration for rapid alert creation, volunteer coordination, and real-time match-alert display.
5. 5. To build a volunteer Android application using Kotlin, CameraX, ML Kit, and TensorFlow Lite that enables real-time crowd scanning for multiple missing children simultaneously.
6. 6. To validate the system's effectiveness against existing government and NGO solutions and demonstrate a meaningful improvement in search capacity over manual, human-only methods.

device face detection and recognition; offline Nearby Connections mesh with custom multi-hop store-and-forward routing; human-verified match confirmation
Geographical scope - Geofenced radius of approximately 2 km around the reporting kiosk Volunteer network - Pre-registered, credible volunteers — police personnel, NCC/NSS cadets, private security staff, shopkeepers, and NGO partners Privacy architecture - All face matching performed on-device; no biometric data uploaded to central servers; alert data is ephemeral and auto-expires

Limitations / Exclusions:

Legal enforcement Does not replace formal police investigation, FIR procedures, or judicial processes Long-term missing-person cases Designed for “golden hour” recovery at a bounded event; not intended for open-ended, long-term missing-person searches Background surveillance The volunteer's camera activates only after an alert is received and the app is opened; there is no continuous or passive crowd surveillance Autonomous declaration The system never autonomously declares a child “found” — a human volunteer must visually confirm every candidate match before it is reported Mass public adoption Targets pre-registered, duty-bound volunteers; the design does not assume spontaneous participation from unregistered festival attendees

Methodology

Research Gap / Novelty:

Existing systems reviewed — TrackChild, Khoya-Paya, ReUnite, Garuda Rakshak, the Maha Kumbh Digital Khoya-Paya Kendras, Project Guardian, and AMBER Alert — each suffer from one or more of the following structural limitations: they are reactive (a photo must be uploaded and matched after the fact), centralised (matching happens on a remote server), internet-dependent (they fail exactly when festival-grade crowds saturate local cellular networks), or centre-/access-bound (restricted to police devices or requiring physical arrival at a fixed location). NextGen Rakshak's novelty lies in combining, for the first time in a single system, four properties that no reviewed solution offers together: Instant, decentralised push of a missing child's photograph and face embedding to every registered volunteer within a geofenced radius, the moment an alert is filed — not after a child is separately found and uploaded. Fully on-device face matching (ML Kit detection + MobileFaceNet/TFLite embedding + cosine similarity) on the volunteer's own smartphone as they move through the crowd, with zero biometric data ever leaving the device or touching a central server. A genuinely offline fallback: a custom multi-hop store-and-forward routing layer (message ID, TTL, duplicate suppression) built on top of Google Nearby Connections, so alerts continue to propagate hop-by-hop even when cellular networks are fully congested — a real, everyday constraint at Indian mass gatherings. A credible, realistic volunteer model that targets pre-registered duty-bearers (police, NCC/NSS cadets, security staff, shopkeepers) rather than assuming spontaneous cooperation from anonymous festival-goers, making the system deployable without depending on mass public adoption.

Proposed Methodology Steps:

1. Data / requirement collection
2. System design / experimental design
3. Development / implementation
4. Offline-mesh and multi-alert integration testing
5. Performance evaluation and benchmarking
6. Field-style demonstration, documentation, and refinement

Detailed Methodology:

Step 1 — Data / Requirement Collection: Gather functional and non-functional requirements from the problem landscape (NCRB data, existing-system limitations, expert commentary on the golden hour). Collect a small labelled face dataset (team members + public frontal-face datasets) for testing detection/embedding accuracy. Define target users (police kiosk operators, volunteers) and their workflows. Step 2 — System Design: Design the three-component hybrid architecture — Police Kiosk Portal (Next.js), Volunteer Android App (Kotlin), and optional Raspberry Pi fixed-camera node — around a Cloud layer (Firebase: Firestore, Auth, Storage, Cloud Functions, FCM), an Edge layer (on-device ML pipeline), and a Mesh layer (Nearby Connections + custom store-and-forward routing). Define data schemas (alert documents, embeddings, match reports), API contracts between components, and the privacy-by-design rules (on-device-only matching, ephemeral alert data, human-in-the-loop confirmation). Step 3 — Development / Implementation: Build the kiosk portal (alert creation, live match dashboard, authentication with role-based claims). Build the volunteer app (CameraX capture, ML Kit face detection with Euler-angle frontal filtering, TensorFlow Lite MobileFaceNet embedding extraction, cosine-similarity matching, match confirmation UI). Implement the offline mesh layer (UUID tagging, TTL-based relay, duplicate suppression) and, if time permits, the Raspberry Pi node (OpenCV + TFLite polling the kiosk API). Step 4 — Offline-Mesh and Multi-Alert Integration Testing: Test alert propagation across 5–10 phones in both internet and Bluetooth/Wi-Fi-Direct-only modes to validate multi-hop relay, TTL expiry, and duplicate suppression. Stress-test the matching pipeline against up to 50 concurrent simulated active alerts to confirm no measurable performance degradation. Step 5 — Performance Evaluation: Benchmark face detection/recognition accuracy, per-face inference latency, alert delivery time (internet vs. offline mesh), and volunteer scan throughput against the target metrics defined in the requirements (see Results section below). Step 6 — Demonstration, Documentation, and Refinement: Run an end-to-end simulated scenario (kiosk alert → volunteer scan → match confirmation → kiosk dispatch), record results, document limitations honestly, and refine thresholds (e.g., the 0.75 cosine-similarity confirmation threshold) based on observed false-positive/false-negative rates.

Literature Review (10 entries)

1	(via ThePrint)	2023	(~6% rise over 2023), 67.1% recovery rate, but resolution typically spans months/years	minute "golden hour" window relevant at a live crowd event
2	TrackChild Portal, Ministry of WCD	2012	Central government database for missing/found children with inter-state coordination	Reactive, post-incident record entry only; no real-time search; fully internet-dependent
3	Khoya-Paya Portal	2015	Citizen-facing portal for reporting missing/found children	Manual search only; no automated face matching; no instant push alerts
4	ReUnite — Bachpan Bachao Andolan + Capgemini (Amazon Rekognition)	2018	Mobile app using cloud-based facial recognition to match uploaded photos of missing children	Requires active internet + centralised cloud processing (privacy concern); reactive "upload-and-wait" model, not proactive scanning
5	Garuda Rakshak — DSP Mutual Fund + Falco Robotics	2025	Drone-based system using ID wristbands and real-time geo-data to locate separated children at Kumbh	Requires physical wristband issuance and drone infrastructure per event; not scalable to smaller, everyday festivals
6	Digital Khoya-Paya Kendra, UP Govt. (Maha Kumbh 2025)	2025	10 AI-enabled fixed centres with facial recognition; ~54,357 people reunited including all 18 reported-missing children	Centralised and centre-bound — a person must physically reach a fixed kendra; no push of the search into the moving crowd via citizen/volunteer devices
7	Project Guardian, Thoothukudi District Police	2025	Police-only smartphone app for instant photo sharing across duty officers; 12 children rescued at Dasara festival	Restricted to police devices only; no automated face detection/recognition; no offline mesh capability
8	AMBER Alert System, US DOJ	2025	National wireless/broadcast alert system; 1,312 child recoveries, 252 via Wireless Emergency Alerts	Network/cellular-infrastructure dependent; passive one-way broadcast; issued for <1% of missing-child cases meeting strict abduction criteria
9	MobileFaceNet (mobile face-recognition architecture)	2018	Lightweight CNN (<1M parameters), >99% accuracy on LFW benchmark, 128-d embeddings, designed for on-device inference	Requires an application-level system (detection pipeline, offline transport, alert logic) to become a deployable end-to-end solution — it is a model, not a system
10	Google Nearby Connections API	2015	Peer-to-peer discovery and data transfer between nearby Android devices via BLE + Wi-Fi Direct, fully offline	Provides only pairwise/cluster connections, not multi-hop mesh routing — reaching a volunteer several hops away requires a custom store-and-forward layer built on top of it

Proposed Assessment Criteria (10 entries)

Dimension	Type	Metric / Indicator	Target / Benchmark	Evidence Source
Technical performance	Quantitative	Face detection accuracy	≥95% on frontal faces	ML Kit detection logs on test dataset
Technical performance	Quantitative	Face recognition accuracy	≥90% on frontal faces under festival-like lighting	On-device match logs vs. ground-truth labelled set
Technical performance	Quantitative	Inference latency per face	<200 ms on a mid-range Android device	Benchmarked timing logs (2–4 GB RAM test devices)
System performance	Quantitative	Alert delivery time (internet mode)	<5 seconds	FCM delivery timestamp logs
System performance	Quantitative	Alert delivery time (offline mesh)	<30 seconds	Multi-hop relay timestamp logs (5–10 phone test)
Scalability	Quantitative	Concurrent active alerts supported	50 alerts with no measurable performance loss	Stress-test logs (frame-processing time vs. alert count)
Usability / field impact	Quantitative	Volunteer scan throughput	200+ faces per hour per volunteer	Simulated crowd-scan trial logs
Privacy compliance	Quantitative	On-device-only processing	Zero biometric data transmitted off-device	Network traffic audit / packet capture during test runs
Reliability	Quantitative	Human-in-the-loop verification	100% of matches visually confirmed before dispatch	Code review + test-case walkthrough of match flow
Comparative benchmark	Quantitative	Search capacity vs. manual kiosk-only method	≥100× improvement over baseline (50–100 faces/hr manual)	Calculated from volunteer network throughput vs. single-kiosk baseline

Resource Requirements (13 items)



Software	firebase (Firestore, Authentication, Cloud Storage, Cloud Functions, Cloud Messaging)	1000	backend, real-time database, auth, alert distribution, serverless embedding computation	Smitraj Bankar
Software	Next.js 14, React 18, TypeScript, Tailwind CSS	0	Police Kiosk Portal web application	Smitraj Bankar
Software	Kotlin, Android SDK, Jetpack Compose, CameraX	0	Volunteer Android application development	Smitraj Bankar
Software	Google ML Kit (Face Detection), TensorFlow Lite, MobileFaceNet (pre-trained model)	0	On-device face detection and embedding extraction	Smitraj Bankar
Software	Google Nearby Connections API (Play Services)	0	Offline peer-to-peer mesh alert relay	Smitraj Bankar
Software	Android Studio, Visual Studio Code, Git/GitHub, Postman, Figma	0	IDEs, version control, API testing, UI/UX wireframing	Smitraj Bankar
Hardware	Android smartphones, 2 GB+ RAM, Android 8.0+	5 Unit	Volunteer app testing, offline mesh multi-hop testing	Smitraj Bankar
Hardware	Laptop / tablet	1 Unit	Police Kiosk Portal test device	Smitraj Bankar
Hardware	Raspberry Pi 4 (4 GB) + Pi Camera Module V2	5000	Fixed-camera edge node at simulated exit gate	Smitraj Bankar
Hardware	USB power bank (5V, 2.5A)	500	Portable power for Raspberry Pi node during demo	Smitraj Bankar
Laboratory	Department Computer / Project Lab with internet access	-	Development, integration testing, and demo rehearsal	Dr. A. B. Pawar (Project Guide)
Dataset	Small self-collected frontal-face dataset (consenting team members) + public benchmark subset (e.g., LFW)	20-50	Testing/benchmarking face detection and recognition accuracy	Smitraj Bankar
Permission	None required for MVP scope — testing confined to team-owned devices and consenting team members	-	N/A (no real missing-child data or live public deployment involved in academic MVP)	-

Ethical, Safety & Permission Requirements

- ✓ **AI-generated content or AI-assisted writing** — AI tools were used to assist with documentation drafting, literature synthesis, and diagram/report formatting during proposal preparation; all technical design and implementation is the team's own work
- ✗ Animal study
- ✓ **Confidential data** — Face images/embeddings are treated as sensitive biometric data even in test form; project explicitly implements privacy-by-design (on-device-only processing, no central biometric storage, auto-expiring alerts) as a core design requirement, not just a compliance afterthought
- ✗ External organization data
- ✗ Hazardous material
- ✓ **Human participants / survey** — Limited to team members who voluntarily provide their own photographs for testing face detection/recognition accuracy — no external human subjects, no real missing children, no minors' data
- ✗ Patient / medical data

Expected Deliverables

#	Deliverable	Details
1	Project report	A comprehensive final project report (expanding on the current synopsis) covering the problem landscape, literature review, system architecture, module-wise functional description, requirements, feasibility assessment, testing/benchmark results, screenshots of the working system, and an honest discussion of limitations — formatted to the department's submission standard.
2	Prototype / model / software / design	A working MVP comprising: (a) the Police Kiosk Portal (Next.js web app) for alert creation and a live match dashboard, and (b) the Volunteer Android App (Kotlin) with on-device face detection (ML Kit), on-device face recognition (MobileFaceNet/TFLite), and an offline multi-hop mesh relay (Nearby Connections), demonstrated end-to-end across 5+ test devices in both internet and fully offline modes. The Raspberry Pi fixed-camera node is included as an optional extension if the timeline permits.
3	Dataset / survey / experimental result	Benchmark results for face detection accuracy, face recognition accuracy, per-face inference latency, alert delivery time (internet mode vs. offline mesh mode), and volunteer scan throughput, measured against the non-functional-requirement targets defined earlier in this proposal, along with the small self-collected/consented test face dataset used to generate them.
4	Research paper / conference paper	A potential submission to an IEEE/Springer conference or a Scopus-indexed journal on one of the following angles arising from this work: (i) optimising an on-device face-detection-and-recognition pipeline for real-time crowd scanning, (ii) performance evaluation of a custom multi-hop store-and-forward mesh built over the Google Nearby Connections API, or (iii) a privacy-by-design framework for crowdsourced public-safety edge-AI systems.
5	Patent filing / patent publication	The realistic IP target is a software copyright registration for the codebase (kiosk portal, volunteer app, and custom mesh-routing implementation), following the same process the team has already used successfully for two prior software copyrights. A formal utility/design patent is not the primary target for this project since it combines existing technologies (Firebase, ML Kit, TensorFlow Lite, Nearby Connections) into a novel architecture rather than claiming a novel underlying algorithm.

Publication / Patent Plans

Output Type	Tentative Title	Venue / Journal	Expected Timeline	Details
Journal Paper	"NextGen Rakshak: A Privacy-Preserving Edge-AI System for Offline-First Lost-Child Recovery at Mass Gatherings Using On-Device Face Matching and Multi-Hop Mesh Networking"	IEEE conference on computing/communication technologies (e.g., IEEE ICCNC or an equivalent Scopus-indexed conference on edge-AI/mobile computing — to be confirmed with project guide closer to submission)	April 2027	-
Patent Filing	"NextGen Rakshak — Offline-First Edge-AI Lost Child Recovery Software System"	Software Copyright — Mobile & Web Application (India, Copyright Office)	May 2027	-

Work Plan & Timeline



MILESTONE	Phase 1	Requirement collection, literature review, and finalization of synopsis (problem landscape, existing-system gap analysis, tech-stack decisions)	2 weeks	11 Aug 2026	Adaptability & Learning Agility Communication Skills Requirement Analysis Time Management
MILESTONE	Phase 2	System design: three-layer architecture (Cloud/Edge/Mesh), Firestore schema, API contracts, UI/UX wireframes — largely completed since project start; logged at the earliest available portal date	2 weeks	12 Aug 2026	Application Design Database Design Project Planning & Execution System Architecture
MILESTONE	Phase 3	Development Sprint 1: Police Kiosk Portal (Next.js + Firebase Auth/Firestore) and Volunteer App skeleton (Kotlin, Jetpack Compose, CameraX, permissions)	2 weeks	13 Aug 2026	Application Development Authentication & Authorization Coding Standards Version Control
MILESTONE	Phase 4	Development Sprint 2: On-device ML pipeline — ML Kit face detection, MobileFaceNet/TFLite embedding extraction, cosine-similarity matching	2 weeks	27 Aug 2026	Application Development Model Evaluation Model Selection Problem Solving
MILESTONE	Phase 5	Development Sprint 3: Offline mesh (Nearby Connections + custom store-and-forward routing — UUID, TTL, duplicate suppression) and optional Raspberry Pi node	1 week	03 Sep 2026	Adaptability & Learning Agility Application Development Problem Solving Secure Coding
MILESTONE	Phase 6	Integration, multi-alert stress testing (up to 50 simulated alerts), 5–10 device offline-mesh testing, and bug fixing	1 week	10 Sep 2026	Deployment & Documentation Teamwork & Collaboration Testing
MILESTONE	Phase 7	Final project report writing and documentation	2 weeks	24 Sep 2026	Communication Skills Documentation Quality Professionalism & Ethics
MILESTONE	Phase 8	Performance optimization: cosine-similarity threshold calibration, inference-latency tuning, battery/throughput optimization based on test data	2 weeks	08 Oct 2026	Application Development Model Evaluation Overall Project Quality Problem Solving
MILESTONE	Phase 9	Deployment: Kiosk Portal hosted on Firebase Hosting, Volunteer APK build for sideload distribution, optional Pi node field setup	2 weeks	22 Oct 2026	Deployment & Documentation Deployment & Inference System Architecture
MILESTONE	Phase 10	Field test / pilot demonstration: controlled crowd-scan trial with 5–10 volunteers at a college event or simulated gathering	3 weeks	12 Nov 2026	Overall Project Quality Presentation Skills Teamwork & Collaboration Testing
MILESTONE	Phase 11	Research paper drafting using field-test and benchmark results; conference submission	3 weeks	03 Dec 2026	Communication Skills Documentation Quality Presentation Skills
SEMESTER BREAK	End of Semester 1 — Marks Evaluation Point	End of Semester 7 — marks for Phase 1–11 milestones calculated at this point Marks for this semester are calculated on milestones completed before this break.	Break period	15 Dec 2026	-
MILESTONE	Phase 12	Patent/copyright filing preparation and submission	4 weeks	15 Jan 2027	Documentation Quality Professionalism & Ethics Project Completion Rate



Email: smitrajbankar11@gmail.com

Responsibility: Overall project lead and system architect — owns the end-to-end architecture (Cloud/Edge/Mesh layers), backend (Firebase Cloud Functions, Firestore schema), on-device ML pipeline integration (ML Kit + MobileFaceNet/TFLite), offline mesh implementation (Nearby Connections + custom routing), cross-component integration, and final report/research paper/patent filing coordination

Mobile: 705/991669

Expected Contribution: ~40% — First Author

Student 2

Name: BHAKARE TANISHKA SHARAD

Email: tanishkabhakare7@gmail.com

Responsibility: Web Portal development — builds the Police Kiosk Portal in Next.js/React/TypeScript/Tailwind CSS, including the alert-creation form, live matches dashboard, and Firebase integration (Auth, Firestore listeners)

PRN / Roll No.: UCS23F1011

Mobile: 9699186099

Expected Contribution: ~20% — Co-Author

Student 3

Name: NARKHEDE ATHARVA ANANTKUMAR

Email: atharvanarkhede969@gmail.com

Responsibility: UI/UX design across both platforms and QA — designs and wireframes (Figma) the Police Kiosk Portal and Volunteer Android App interfaces, and leads functional/integration testing across the system (multi-device offline mesh tests, multi-alert stress tests, UI/UX validation)

PRN / Roll No.: UCS23M1094

Mobile: 7588343686

Expected Contribution: ~20% — Co-Author

Student 4

Name: DHADGE VEDANT SANJAY

Email: dhadgevedant@gmail.com

Responsibility: Android application development and Firebase integration — builds the Volunteer Android App (Kotlin, Jetpack Compose, CameraX) and wires it to Firebase (Authentication, Firestore sync, Cloud Messaging) for alert receipt and match reporting

PRN / Roll No.: UCS23M1034

Mobile: 9322260485

Expected Contribution: ~20% — Co-Author

Workflow History

→ **Draft → Pending Guide Review** 0 seconds ago
by BANKAR SMITRAJ DINKAR
Proposal submitted.

→ **Draft (initialized)** 5 days ago
Workflow initialized.